UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 002 · Nernst equation and its derivation · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S002·4	Deck 002, slide 4 (rendered .pptx)
T001 00:21:42	Class 001 AV-summary transcript (covers classes 1 and 2), segment start
PDF p.13	Consolidated Unit 2 slide PDF (older syllabus), page 13
ESA Jan-25 s1 2a	Past paper: Jan 2025, set 1, question 2.a

Contents

1. 1 · Free energy and electrode potential
2. 2 · Derivation of the Nernst equation for a single electrode
3. 3 · Using the single-electrode equation: concentration and temperature
4. 4 · Nernst equation for a cell
5. Past-paper questions

1 Free energy and electrode potential

Definitions

Maximum work. The maximum work that can be obtained from a system equals the decrease in free energy: $-\Delta G = W_{max}$. For an electrochemical system, W_{max} = total charge available × energy available per unit charge. S002·3 · T001 00:19:38 · T001 00:20:11

Faraday constant, F. The charge carried by one mole of electrons, 96500 C mol^{-1} on the slides. S002·3 · T001 00:20:21

Statements in the class material

> *"The maximum work that can be obtained is $-\Delta G = W_{max}$. For an electrochemical system, maximum work done is total charge available × energy available per unit charge. Therefore $W_{max} = nFE$; $\Delta G = -nFE$. Under standard conditions, $\Delta G^\circ = -nFE^\circ$."* S002·3

Key points

1. Total charge available = moles of electrons exchanged (n) × charge per mole of electrons (F) = nF; energy per unit charge = electrode potential E (volt = joule per coulomb). ESA ★★★ S002·3 · T001 00:20:21 · T001 00:20:50
2. $\Delta G = -nFE$: for a spontaneous reaction $\Delta G < 0$, so $E_{cell} > 0$; if $E_{cell} < 0$ the reaction is non-spontaneous. ESA ★★★ T001 00:16:39 · T001 00:16:51 · PDF p.10
3. E_{cell} is the driving force for the cell reaction. ESA ★★ T001 00:16:15 · PDF p.10

Why?

Why is $W_{max} = nFE$?
Work = charge × potential difference. n moles of electrons carry nF coulombs, and each coulomb gains E joules (1 V = 1 J C^{-1}).
S002·3 · T001 00:20:11

2 Derivation of the Nernst equation for a single electrode

Definitions

Nernst equation. A quantitative relationship between electrode potential and concentration of species with which the electrode is reversible. S002·3 · T001 00:19:17

Reaction quotient, Q. The concentration of the products divided by the concentration of the reactants, each raised to its stoichiometric coefficient, at any instant (not only at equilibrium). S002·4 · T001 00:22:05 · T001 00:24:15

Standard electrode potential, E°. The electrode potential when all species are at unit concentration (1 M), gases at 1 atm, at 298 K. S002·4 · T001 00:19:00

Statements in the class material

> *"A thermodynamic equation which relates reaction quotient and decrease in free energy is given by $\Delta G = \Delta G^\circ + RT\ln Q$, where Q is the reaction quotient. The reaction quotient for the reaction is $Q = [\mathrm{M}]/[\mathrm{M}^{n+}]$."* S002·4

Key points

1. The electrode reaction is written as a **reduction**, $\mathrm{M}^{n+} + ne^- \rightleftharpoons \mathrm{M}$, because electrode potentials are reduction potentials. ESA ★★★ S002·3 · T001 00:19:38
2. The system is a metal rod dipped in a solution of its own ions. ESA ★★ T001 00:19:56
3. Electrochemical series values are standard (E°: 1 M, 1 atm, 298 K), but real systems are seldom standard; the Nernst equation corrects E° for concentration, temperature and pressure. ESA ★★ T001 00:19:00 · T001 00:19:25
4. $[\mathrm{M}] = 1$ because the metal is a pure substance (pure solids have unit activity). ESA ★★★ S002·5 · T001 00:23:17
5. Natural log to common log: $\ln x = 2.303 \log x$; at 298 K, $2.303RT/F = 0.0591$ V. ESA ★★★ S002·5 · T001 00:23:29 · T001 00:23:50

Why?

Why does the sign change from $-\frac{RT}{nF}\ln\frac{[\mathrm{M}]}{[\mathrm{M}^{n+}]}$ to $+\frac{RT}{nF}\ln[\mathrm{M}^{n+}]$?
With $[\mathrm{M}] = 1$, $\ln\frac{1}{[\mathrm{M}^{n+}]} = -\ln[\mathrm{M}^{n+}]$, and the two minus signs make a plus.
S002·4 · S002·5

Why is the derivation done for the reduction reaction?
Standard electrode potentials are tabulated as reduction potentials, so writing $\mathrm{M}^{n+} + ne^- \to \mathrm{M}$ gives E on the same scale as E°.
T001 00:19:38

Exam traps

Exam trap. While extending to the cell the lecturer says "$\Delta G = \Delta G^\circ + \frac{RT}{nF}\ln Q$" (T001 00:24:39). That is wrong: the free-energy equation has $RT\ln Q$; the $1/nF$ appears only after dividing by $-nF$. ADDED FOR CLARITY T001 00:24:39 · S002·4

Exam trap. Sign: the single-electrode (reduction) form has $+\frac{0.0591}{n}\log[\mathrm{M^{n+}}]$; the cell form has $-\frac{0.0591}{n}\log Q$. Both are the same equation, since $Q = 1/[\mathrm{M^{n+}}]$ for the electrode. ADDED FOR CLARITY S002·5

3 Using the single-electrode equation: concentration and temperature

Key points

1. Raising $[\mathrm{M}^{n+}]$ raises E (the reduction becomes more favourable); diluting lowers it. Each tenfold change shifts E by $0.0591/n$ V. ESA ★★ ADDED FOR CLARITY S002·5
2. At $[\mathrm{M}^{n+}] = 1$ M, $\log 1 = 0$ and $E = E^{\circ}$: the Nernst equation reduces to the standard potential. ESA ★★ ADDED FOR CLARITY S002·5
3. The 0.0591 is valid only at 298 K. At another temperature use $2.303RT/F$ with that T (0.0611 V at 308 K). ESA ★★ S002·5 · T001 00:25:37

Exam traps

Exam trap. Do not use 0.0591 with $\ln$: 0.0591 already contains the 2.303 and goes with $\log_{10}$; with $\ln$ use $RT/nF = 0.0257/n$. ADDED FOR CLARITY S002·5

4 Nernst equation for a cell

Definitions

Standard emf, E°_{cell}. The emf of the cell when all concentrations are 1 M, all partial pressures 1 atm and the temperature 298 K. S002·5 · PDF p.10

Statements in the class material

> *"Nernst equation may also be used to calculate emf of electrochemical cells. For the cell reaction $aA + bB \rightleftharpoons cC + dD$, the Nernst equation is $E = E^\circ - \frac{0.0591}{n}\log\frac{[C]^c[D]^d}{[A]^a[B]^b}$; n = no. of electrons transferred, E°_{cell} = std. emf of the cell."* S002·5

Key points

1. For a cell only Q changes; $\Delta G = -nFE_{cell}$ and $\Delta G^\circ = -nFE^\circ_{cell}$ are substituted exactly as for one electrode. ESA ★★★ T001 00:24:15 · T001 00:24:52
2. n is the number of electrons in the **balanced** overall reaction: 2Al + $3Cu^{2+}$ exchanges 6 electrons. ESA ★★★ (added for clarity) ESA Jan-25 s1 2a
3. Solids (Al, Cu, Zn) do not appear in Q; only dissolved ions (and gas pressures) do. ESA ★★★ S002·5 · ESA Jan-25 s1 2a
4. The cell form gives the same emf as $E_{cathode} - E_{anode}$ with each electrode from E2.5 (check: Daniell 1.1296 V both ways). ESA ★★ (added for clarity) T001 00:16:00

Why?

Why is the cell equation written with a minus sign and Q = products/reactants?
It comes straight from $\Delta G = \Delta G^\circ + RT\ln Q$: dividing $-nFE = -nFE^\circ + RT\ln Q$ by $-nF$ gives $E = E^\circ - \frac{RT}{nF}\ln Q$. More products (larger Q) lower the emf.
S002·5 · T001 00:24:52

Exam traps

Exam trap. ESA Jan 2025 prints "2Al + $3Cu^{2+}$ → $2Al^{3+}$ + Cu". Balance it (3Cu) and take $n = 6$; using $n = 2$ or $n = 3$ gives a wrong emf (2.0049 V for $n = 3$). (added for clarity) ESA Jan-25 s1 2a

Exam trap. Raise each concentration to its coefficient: $[Al^{3+}]^2/[Cu^{2+}]^3$, not $0.35/0.6$.
(added for clarity) S002·5 · ESA Jan-25 s1 2a

★ Past-paper questions

Ten deduplicated Chemistry ESA papers (UE22CY151B Jul 2023 ×2, Dec 2023, May 2024; UE23CY151A/B Jan 2024 set 2, Jun 2024 ×2, Dec 2024, Dec 2024 UE22; UE24CY151A Jan 2025 sets 1 and 2) were checked. The Nernst derivation is asked directly twice: ESA Dec 2024 Q2.a (single electrode, 5 marks) and ESA Jan 2025 set 1 Q2.a (cell form plus emf, 6 marks). In addition, every UE23/UE24 paper has a Q2 cell numerical (calomel, redox, concentration cell) that uses the cell form of the equation; those are mapped to classes 004–006. The equilibrium constant, the temperature effect and the E vs log[M^{n+}] plot were never asked.

2 · Derivation of the Nernst equation for a single electrode

ESA Dec 2024 · 5 marks · UE23CY151A/B set 1, Q2.a · solved in W2.1

Derive the Nernst equation for a single electrode undergoing the following electrode reaction: $M^{n+} + ne \rightarrow M$.

4 · Nernst equation for a cell

ESA Jan 2025 · 6 marks · UE24CY151A set 1, Q2.a · solved in W4.1

Derive Nernst equation for the given cell and calculate the emf of the cell. $2Al + 3Cu^{2+}(0.6\,M) \rightarrow 2Al^{3+}(0.35\,M) + Cu$ (Given: $E^{\circ}_{cell} = 2.0$ V).